CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

I Semester of BSc Biological Sciences Examination March 2018

BS 109 Chemistry

Date: 27/03/2018 Day: Tuesday Time: 10.00 am to 10.30 am Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions. [20]

1 __________ is measure of randomness.

a) Enthalpy b) Entropy c) Energy d) Heat

2 For the reaction, if the value of rate constant is 0.023 Mol^{-1} lit^{1} s^{-1} then what is the order of reaction?

a) First b) Second c) Third d) Zero

3 The change in concentration of reactant or product with time is called ……….

a) order of reaction b) Rate of reaction c) Rate constant d) Molecularity

4 For reaction to non-spontaneous, the value of ΔG is ……..

a) Negative b) Positive c) Zero d) Both Positive and Negative

5 What is the relation between work and force?

a) W = f/r b) W = f – r c) W = f × r d) f = W – r

6 For reaction to feasible, the value of ΔG is …….

a) Negative b) Positive c) Zero d) Both Positive and Negative

7 $AlCl_3$ is ……….

a) Electrophile b) Nucleophile

c) Neutral electrophile d) positively charged electrophile

8 Carbon has unique property of linking itself to other carbon atoms to give open chain or/and cyclic structure. This property of self-linking is termed ……….

a) polymerization b) oligomerization

c) catenation d) none of above

9 The stereoisomer which is non-super imposable on its mirror image is known as ……

a) Enantiomers b) Diastereomers

c) Homomers d) Tautomers

10 A tetrahedral atom attach to four different substituent is known as ………

a) Asymmetric center b) Chirogenic center

c) Chiral center d) all of these

11 The classification of the elements in modern periodic table has been made on the basis of ……….

a) electronic configuration of the element

b) electronic configuration of valence shell of the element

c) atomic weight of the element

d) atomic size of the elements

12 Which repulsion between electron-pairs is strongest one?

a) lone pair – lone pair

b) lone pair – bond pair

c) bond pair – bond pair

d) none of these

13 The orbital interaction between the π system and the adjacent σ bonds of the substituents group(s) in an organic molecule is described by ……….

a) Hyper conjugation

b) Mesomeric effect

c) Electromeric effect

d) Inductive effect

14 How many lone pairs are present on nitrogen atom of ammonia molecule?

a) 0 b) 1 c) 2 d) 3

15 Transition elements in periodic table are also known as ………..

a) d – block elements

b) s – block elements

c) p – block elements

d) f – block elements

16 Ionization energy values are expressed in ……….

a) electron volts per atom (eV/atom)

b) kilo calories per mole (kcal/mole)

c) kilo joules per mole (kJ/mole)

d) all of above

17 Hybridization of B – atom in BF_3 molecule is ……….

a) sp b) sp^2 c) sp^3 d) dsp^3

18 Value of bond angle in CH_4 ___________.

a) 90° & 120° b) 90° & 180°

c) 109° 28’ d) 90°

19 From the following which molecule(s) have bon order zero?

a) H_2 b) He_2 c) O_2 d) N_2

20 “Electromeric effect is a temporary effect which comes into play instantaneously on the requirements of the attacking reagent.”

a) True

b) False

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

I Semester of BSc Biological Sciences Examination March 2018

BS 109 Chemistry

Date: 27/03/2018 Day: Tuesday Time: 10.31 am to 01.00 pm Maximum Marks: 50

Important Instructions:

- **Section I and II must be attempted in TWO ANSWER SHEET.**
- Make suitable assumptions and draw neat figures wherever required.
- Use of non-programmable calculator is allowed.
- Show necessary calculations.

SECTION -I **[20]**

Q - II

[A] Discuss the structures of XeF_4 and H_2O molecules with the help of VSEPR theory. **[04]**

OR

[A] Define the term "Ionization energy". Discuss the factors that affect the magnitude of ionization energy. **[04]**

[B] Define the term hybridization. Discuss the hybridization of C – atom in CH_4 molecule. **[03]**

[C] Explain Lowery Bronsted concept of acid and base by giving suitable examples. **[03]**

Q - III

[A] Define the following: **[04]**

i) Electron affinity

ii) Electronegativity

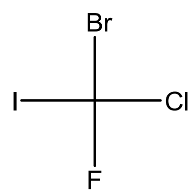
iii) Modern periodic law

iv) Mendeleev's periodic law

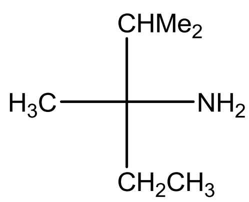
[B] Construct the molecular orbital diagram of N_2 (N – atom, Z = 7) molecule and calculate the bond order. **[03]**

[C] Give R and S configurations of following chiral organic molecules. **[03]**

i)



ii)



OR

[C] Define the term Electron Affinity. List the factors that affect the magnitude of electron affinity. **[03]**

SECTION – II [30]

Q - IV

[A] Explain Mesomeric effect in detail by giving various examples. **[04]**

[B] Explain Inductive effect by giving suitable examples. **[03]**

[C] Explain "All chirogenic centers are stereogenic center but reverse is not true." **[03]**

OR

[C] Discuss the limitation and usefulness of Fischer projection. **[03]**

Q - V

[A] What is heat capacity? Derive the equation for heat capacity at constant pressure and constant volume. **[04]**

[B] Obtain the relation for the rate constant k and activation energy **[03]**

[C] Discuss the Hess's law with suitable example. **[03]**

OR

[C] For reaction $C_2H_5 + OH^- \longrightarrow C_2H_5OH + I^-$ **[03]**

$k = 5.03 \times 10^{-2}\ M^{-1}.\ sec^{-1}$ at 289 K and $k = 6.71\ M^{-1}.\ sec^{-1}$ at 333 K.

What is activation energy of the reaction?

Q - VI

[A] Derive the equation for first order integration rate law and discuss its characteristics. **[04]**

[B] Draw the lines for positive and negative slopes. **[03]**

[C] "O_2 is paramagnetic while O_2^{2-} is diamagnetic" Explain on the basis of Molecular Orbital Theory. (Given: Atomic Number of Oxygen is 8) **[03]**

OR

[C] The rate constant of one reaction is $1 \times 10^{-4}\ min^{-1}$ at 27 ^oC and $2 \times 10^{-4}\ min^{-1}$ at 37 ^oC. Calculate the activation energy of this reaction and its rate constant at 47 ^oC. (R = 1.987 cal.) **[03]**